

NANO MUDHALVAN PROJECT

FITBUDDY AI

AI Fitness Plan Generator

Complete Project Explanation & Development Phases

An AI-based fitness planning application that generates personalized workout plans from user information, goals, experience, available time and equipment.

Project Type: Artificial Intelligence / Web Application

Core Output: Personalized weekly fitness plan

1. Project Overview

FitBuddy AI – Fitness Plan Generator is an AI-based fitness planning application. Instead of providing the same workout routine to everyone, it uses a user's profile and fitness goal to generate a structured starting plan.

The system can collect age, gender, height, weight, fitness goal, experience level, activity level, available days, workout duration, location and equipment. A recommendation engine then selects suitable exercises and organizes them into a weekly schedule.

The application is an educational prototype and should not replace a doctor, physiotherapist or certified fitness professional, especially for injuries or medical conditions.

2. Problem Statement

- Beginners may not know which exercises are appropriate for their experience level.
- Generic workout plans may not match a user's goal, time or equipment.
- Creating a weekly workout schedule manually can be difficult for beginners.
- Users need a simple digital tool for organizing a basic personalized fitness routine.

3. Proposed Solution

FitBuddy AI provides one platform where the user enters basic fitness information and receives a personalized workout plan. The recommendation engine analyzes the inputs, filters suitable exercises and creates a weekly structure.

The plan can include warm-up, exercises, sets, repetitions or duration, rest periods, cool-down and weekly scheduling. Users can update their profile and regenerate the plan.

4. Objectives

- Create an easy-to-use fitness plan generator.
- Collect and store basic user fitness information.
- Generate personalized workout recommendations.
- Match exercise difficulty with the user's experience level.
- Create a weekly workout schedule.
- Allow plan regeneration when user preferences change.
- Demonstrate practical use of AI/recommendation technology.

5. Main Features

Feature	Description
Registration/Login	Creates an account and provides secure access.
Fitness Profile	Stores basic profile and fitness information.
Goal Selection	Allows the user to choose a general fitness goal.

Feature	Description
AI Plan Generator	Processes inputs and creates a personalized plan.
Weekly Schedule	Organizes workouts across selected days.
Exercise Details	Shows exercise, sets, reps/duration and rest.
Regenerate Plan	Creates a new plan after profile/goal changes.
Progress Tracking	Records completed workouts and basic progress.
Admin Module	Manages exercise categories and application data.

6. Input Parameters

- **Age:** Basic personalization parameter.
- **Height and Weight:** Can be used to calculate BMI as a general reference.
- **Fitness Goal:** Examples include general fitness, strength, endurance and weight management.
- **Experience:** Beginner, intermediate or advanced.
- **Activity Level:** Low, moderate or high.
- **Location:** Home or gym.
- **Available Days:** Number of workout days per week.
- **Workout Duration:** Approximate time per session.
- **Equipment:** Bodyweight, dumbbells, bands or gym equipment.

7. System Workflow

1. User opens FitBuddy AI.
2. User registers/logs in.
3. User enters fitness profile and preferences.
4. System validates the input.
5. Basic derived values can be calculated if required.
6. Recommendation engine analyzes goal, level, time and equipment.
7. Suitable exercises are selected from the database.
8. Exercises are arranged into a weekly plan.
9. Plan is displayed and saved.
10. User can update, follow or regenerate the plan.

8. System Architecture

- **Presentation Layer:** Web/mobile screens for login, profile and plan display.
- **Application Layer:** Authentication, validation, business logic and API handling.
- **AI/Recommendation Layer:** Rule-based, scoring-based, ML or LLM-assisted recommendations.
- **Database Layer:** Users, exercises, goals, plans and progress.
- **Security Layer:** Authentication, authorization, validation and data protection.

9. Major Modules

Module	Responsibilities
Authentication	Register, login, logout and session management.
User Profile	Create and update fitness information.
Exercise Database	Store exercise name, category, difficulty, equipment and instructions.
AI Recommendation	Analyze user inputs and select compatible exercises.
Plan Generator	Create daily and weekly workout schedules.
Progress Tracker	Record completed workouts and basic progress.
Admin	Manage exercise data and application content.

10. How AI Works

Rule-Based: Predefined rules select exercises based on goal, level and equipment. This is simple and suitable for a student prototype.

Scoring-Based: Exercises receive compatibility scores based on goal, difficulty, equipment and available time. Higher-scoring exercises are selected.

Machine Learning: A model can learn from a suitable dataset of user profiles and exercise recommendations. This requires data and model evaluation.

Generative AI: An LLM can convert structured recommendations into readable plans, but generated output should be constrained and validated.

Example: Strength goal + beginner + dumbbells → filter beginner strength exercises → select compatible exercises → divide them across chosen days → assign workout parameters → display plan.

11. Suggested Database Design

Table	Important Fields
Users	user_id, name, email, password_hash, created_at
Fitness_Profile	profile_id, user_id, age, height, weight, activity_level, experience
Goals	goal_id, goal_name, description
Exercises	exercise_id, name, category, difficulty, equipment, instructions
Workout_Plans	plan_id, user_id, goal_id, created_at, duration
Plan_Details	detail_id, plan_id, exercise_id, day, sets, reps, rest
Progress	progress_id, user_id, plan_id, date, completed, notes

12. Suggested Technology Stack

- **Frontend:** HTML, CSS, JavaScript, Bootstrap or React.
- **Backend:** Python Flask/FastAPI, Node.js/Express or Java Spring Boot.

- **Database:** MySQL, PostgreSQL, SQLite or MongoDB.
- **AI:** Python recommendation logic, scikit-learn, an ML model or an LLM API.
- **Tools:** VS Code/IDE, Git/GitHub, Postman and browser developer tools.

For a student prototype, HTML/CSS/JavaScript + Python Flask + SQLite/MySQL + Python recommendation logic is a practical combination.

13. Project Development Phases

Phase	Activities	Output
1 – Requirement Analysis	Identify problem, users, inputs, outputs and requirements.	SRS / requirement list
2 – System Design	Design architecture, modules, database and AI logic.	Architecture + diagrams
3 – UI/UX Design	Design login, profile, dashboard and plan screens.	UI prototype
4 – Database Development	Create tables, relationships and CRUD operations.	Working database
5 – Backend Development	Implement authentication, validation, APIs and business logic.	Backend/API
6 – AI/Recommendation	Implement rules, scoring, ML or AI integration.	Plan engine
7 – Integration	Connect frontend, backend, database and AI engine.	Integrated application
8 – Testing	Test functionality, validation, AI output, database and security.	Test report
9 – Deployment	Configure server/database and publish prototype.	Deployed system
10 – Documentation	Prepare report, screenshots, diagrams and presentation.	Final documentation

14. Phase-by-Phase Explanation for Viva

Phase 1 – Requirement Analysis: We identify the problem and decide what information the user enters and what output the application must produce.

Phase 2 – System Design: We design the architecture, database, module interaction and screen flow before coding.

Phase 3 – UI/UX Design: We create simple screens for profile input and plan display.

Phase 4 – Database Development: We create tables for users, profiles, exercises, plans and progress and define relationships.

Phase 5 – Backend Development: We implement login, validation, profile management, APIs and database communication.

Phase 6 – AI/Recommendation: The user's information is processed and compatible exercises are selected and organized.

Phase 7 – Integration: Frontend, backend, database and recommendation engine are connected.

Phase 8 – Testing: We test valid/invalid inputs, login, database operations, plan generation and different profiles.

Phase 9 – Deployment: The completed application is hosted or demonstrated on the local system.

Phase 10 – Documentation: We prepare the report, screenshots, diagrams, test cases, results and presentation.

15. Basic Plan Generation Algorithm

1. Start.
2. Read user profile and preferences.
3. Validate required inputs.
4. Calculate basic derived values if required.
5. Identify goal and experience level.
6. Filter exercises by goal, difficulty and equipment.
7. Select exercises for each workout day.
8. Assign sets, repetitions/duration and rest using the selected training template.
9. Store the generated plan.
10. Display the weekly plan.
11. Allow save/update/regeneration.
12. Stop.

16. Sample Output Format

Day	Workout	Example Structure
Monday	Upper Body	Warm-up → Push → Pull → Shoulder → Cool-down
Tuesday	Lower Body	Warm-up → Squat pattern → Leg exercise → Calf → Cool-down
Wednesday	Recovery	Light activity / mobility / rest
Thursday	Upper Body	Warm-up → Push → Pull → Arms → Cool-down
Friday	Lower Body + Core	Warm-up → Legs → Core → Cool-down
Saturday	Optional Activity	Walking/cardio/mobility depending on plan
Sunday	Rest	Recovery

This table is only an application-output example. Exact exercise volume should be validated for the intended user group.

17. Testing Plan

Test Case	Input/Action	Expected Result
Login	Valid credentials	Dashboard opens
Invalid Login	Wrong password	Error message
Profile	Valid fitness data	Profile saved
Validation	Missing required field	Validation message

Test Case	Input/Action	Expected Result
Goal	Select goal	Goal stored
Plan Generation	Complete profile	Workout plan generated
Equipment	Bodyweight only	Compatible exercises shown
Regeneration	Change goal/level	New plan generated
Database	Save plan	Plan retrievable
Security	Unauthorized request	Access denied

18. Advantages

- Personalized output based on user inputs.
- Saves time in preparing a basic workout schedule.
- Easy digital access.
- Can support different goals and experience levels.
- Can be expanded with progress tracking and notifications.
- Demonstrates a practical AI/recommendation application.

19. Limitations

- Recommendation quality depends on the exercise database and algorithm.
- A rule-based system is not equivalent to professional coaching.
- AI-generated text can be unsuitable if it is not validated.
- The application should not diagnose medical conditions or prescribe rehabilitation.
- Real personalization requires more information and validation than a student prototype may collect.

20. Security and Privacy

- Hash passwords securely; never store plain-text passwords.
- Validate and sanitize all user input.
- Use authentication and authorization for protected pages.
- Protect API endpoints and database credentials.
- Collect only information necessary for the application.
- Avoid exposing private user data in logs or error messages.

21. Future Enhancements

- Mobile application version.
- Wearable-device integration.
- Exercise videos or animations.
- Progress charts and personalized dashboards.

- Voice-based fitness assistant.
- Adaptive plans based on workout history.
- Tamil and English multilingual support.
- Improved recommendation models trained and evaluated on suitable datasets.
- Workout reminders and notifications.

22. Common Viva Questions and Answers

Q1. What is FitBuddy AI?

It is an AI-based fitness plan generator that creates a personalized workout schedule using user fitness information and preferences.

Q2. Why did you choose this project?

It addresses a practical fitness-planning problem and demonstrates how software and AI can provide personalized recommendations.

Q3. What is the role of AI?

AI/recommendation logic analyzes user inputs and helps select and organize compatible exercises.

Q4. What are the main inputs?

Age, height, weight, activity level, experience, goal, available days, duration and equipment.

Q5. What is the output?

A structured workout plan containing workout days, exercises and parameters such as sets, repetitions/duration and rest.

Q6. What database tables are used?

Users, fitness profile, goals, exercises, workout plans, plan details and progress.

Q7. What happens if the user changes the goal?

The system can regenerate a plan using the updated goal and existing profile information.

Q8. What is the future scope?

Mobile apps, wearables, progress analytics, multilingual support and adaptive recommendations.

23. Recommended Presentation Flow

- Introduction and problem statement
- Existing problem and limitations of generic plans
- Proposed FitBuddy AI solution
- Objectives
- System architecture
- Modules
- AI/recommendation workflow
- Database design
- Screens and live demonstration
- Testing and results

- Limitations and future scope
- Conclusion

24. Conclusion

FitBuddy AI – Fitness Plan Generator is a practical project demonstrating requirement analysis, UI design, database management, backend development, recommendation logic, testing and deployment. It converts user fitness preferences into a structured workout plan and demonstrates a real-world application of AI.

For the Nano Mudhalvan demonstration, show one complete flow: enter profile → select goal → generate plan → display weekly schedule → change an input → regenerate → show progress tracking. This clearly demonstrates the end-to-end system.